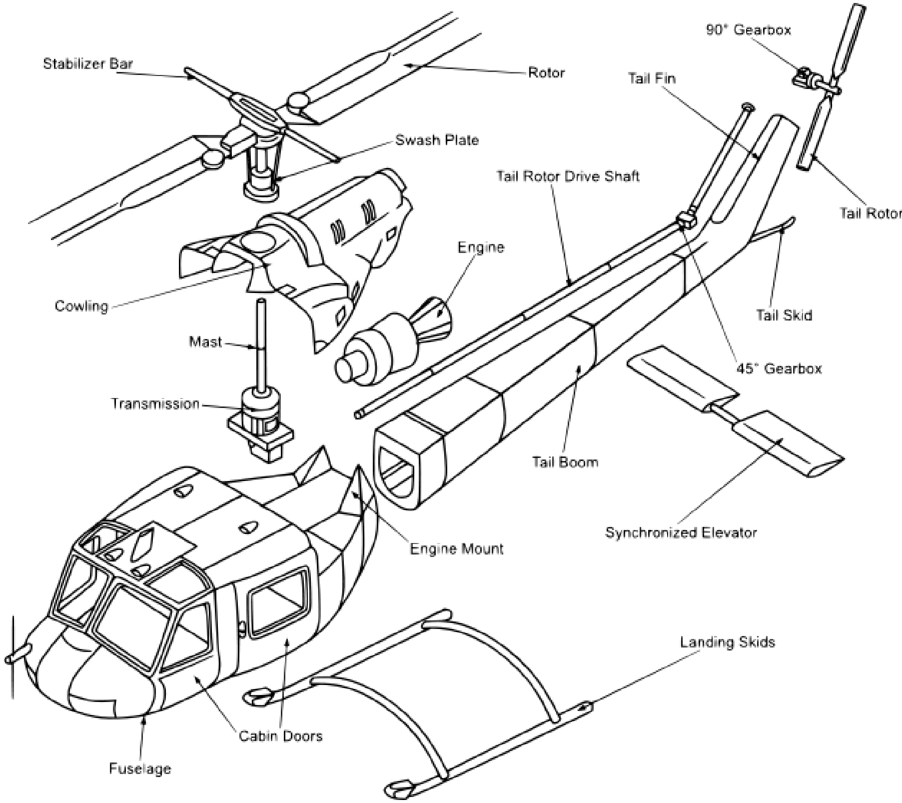
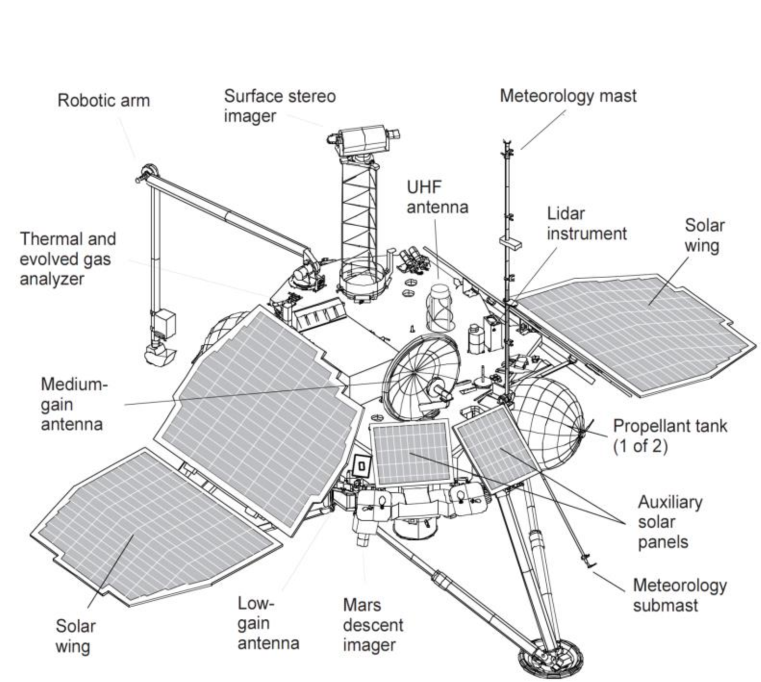
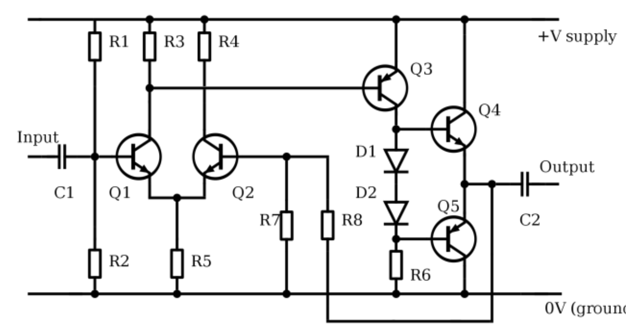




HTMAA Exam
Andrés Faíña (anfv@itu.dk)



Exam

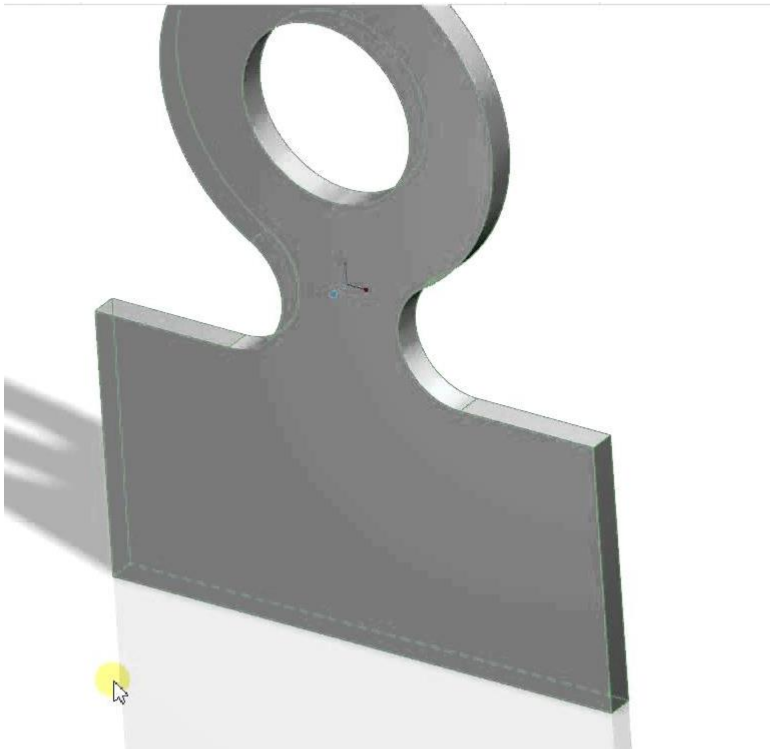
15 minutes group presentation + questions
- all students should present and answer

For each student:
- 15/10 minutes for questions (about course's topics)
  - 3 sections to discuss:
    - Q1: Manufacturing of parts
    - Q2: How a machine works
    - Q3: Electronics

5 min voting and feedback

If possible, bring the prototype ~~connect your smartphone as an extra device to the exam to be able to show the prototype easily.~~

Q1- How would you manufacture these parts?



Q1- How would you manufacture these parts?

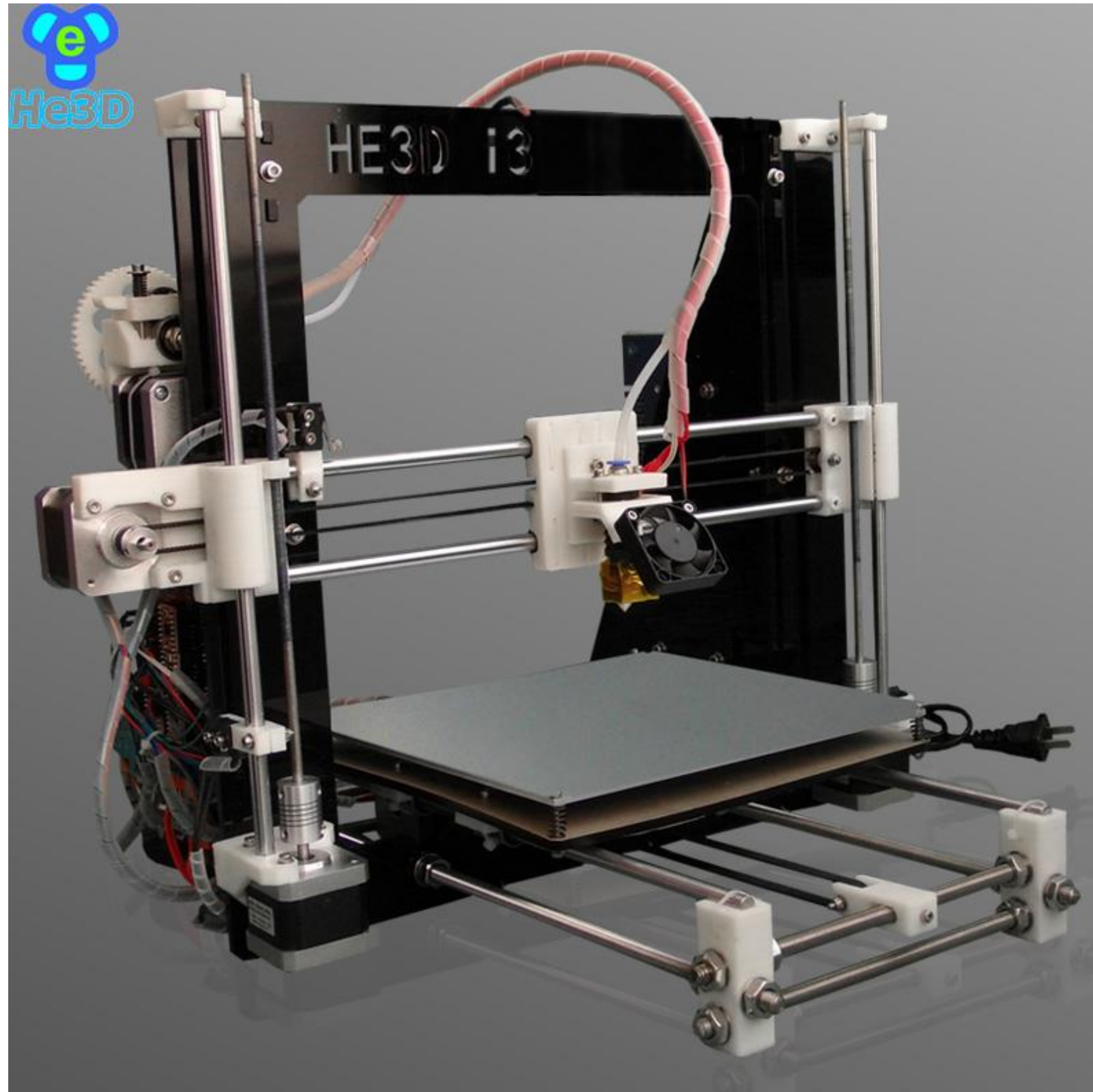
And if you need to make it of metal?

And if you need to make 1000 parts?

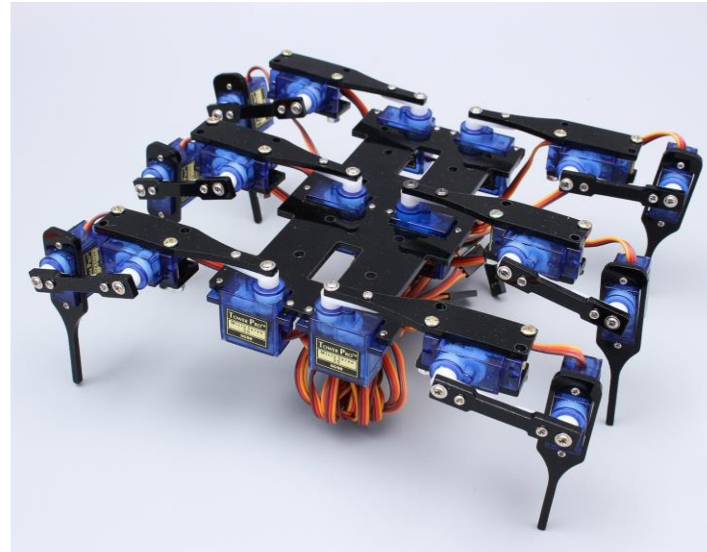
Q2 - How does it work?

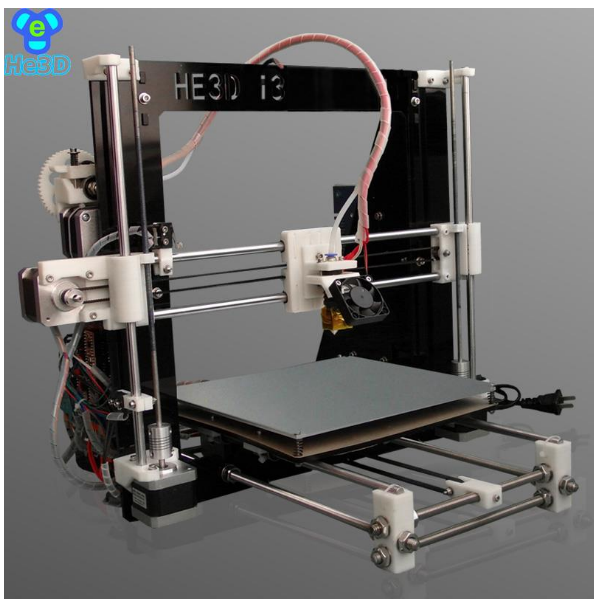
Tips

1. Take your time to analyse the machine
2. Identify components:
    - Linear guides
    - Motors (which type)
    - Bearings
    - Belts, Pulleys, Gears
    - Structural elements
3. Then, try to analyse how it works
    - Start with the motor and describe how it transmits its torque to the end effector
4. Enumerate the DOFs
5. Sensors (in the image or because it may need one)



Q2 - How does it work?





Sensors?

Actuators:

Is it a servo, a DC motor or a stepper motor

WHY?

Mechanisms:

Is there a reduction? **Why?**

Drivers?

Microcontroller?

Q3 - Electronics

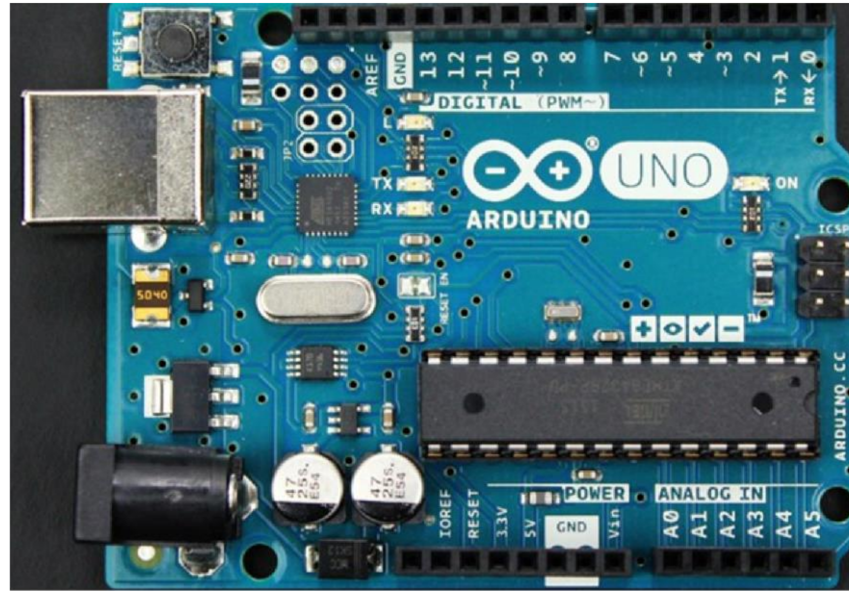
How would you read a switch?

How would you read a potentiometer?

How do you control a DC motor?

How does a servo motor work? (enumerate its elements and describe what they do)

Q3 - Electronics: How do you wire a button?



Please, wire these components

(Maybe there will be a breadboard o multimeter)

Q3 - Electronics

Learn how to wire (and draw schematics) of all the components that we have used:

Switches, LEDs, IR sensors, DC motors, Stepper motors, Servos, etc.

In my experience, half of the students are not able to wire the previous slide with the switch button.

So, practice, use the simulator and check the MAs.

Ready?

Good luck!

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